

MOGA Study Report

Evolutionary Multi-Objective Optimization

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Abstract

This report presents the implementation, evaluation, and extension of a Multi-Objective Genetic Algorithm (MOGA) on five standard ZDT benchmark problems: ZDT1, ZDT2, ZDT3, ZDT4, and ZDT6. Experiments cover D=10 and D=30 under a maximum budget of 200,000 function evaluations. The baseline MOGA is compared with tournament/elitist/archive MOGA, crowding-distance MOGA, crowding plus epsilon-grid archive control, and crowding plus hypervolume-contribution archive pruning. Archive-based and diversity-aware extensions substantially improve convergence and Pareto-front coverage over the baseline on most tested cases, while ZDT4 remains the most difficult benchmark because of its multimodal search landscape.

1. Experimental Setup

Item	Setting
Benchmark problems	ZDT1, ZDT2, ZDT3, ZDT4, ZDT6
Dimensions	D=10 and D=30
Evaluation budget	Maximum 200,000 function evaluations
Independent runs	10 runs per problem, dimension, and algorithm setting
Representation	Binary-coded chromosome representation
Primary metrics	Generational Distance (GD) and Inverted Generational Distance (IGD)
Supporting metrics	Spacing, 2D hypervolume, front size, runtime, hyperparameter sensitivity

Lower GD and IGD are better. GD measures closeness to the reference Pareto front, while IGD emphasizes how completely the reference front is covered. The fixed evaluation budget prevents an improved method from quietly buying better numbers with additional search effort.

2. Algorithms Compared

Algorithm	Description
MOGA	Baseline binary-coded MOGA with roulette-style selection, uniform crossover, bit mutation, and dominance-based ranking.
MOGA Bonus	Tournament selection, elitism, adaptive mutation, and an external nondominated archive.
Crowding	NSGA-II-style crowding distance in selection and archive pruning.
Crowding + epsilon	Crowding with epsilon-grid filtering to control near-duplicate archive points.
Crowding + HV	Crowding with hypervolume-contribution archive pruning.

3. Baseline MOGA Results

GD at D=10

Problem	MOGA
ZDT1	0.22163
ZDT2	0.36434
ZDT3	0.19688
ZDT4	26.8802
ZDT6	1.56325

IGD at D=10

Problem	MOGA
ZDT1	0.01539
ZDT2	0.03393
ZDT3	0.01357
ZDT4	1.68026
ZDT6	0.20628

GD at D=30

Problem	MOGA
ZDT1	0.33038
ZDT2	0.7451
ZDT3	0.31074
ZDT4	104.99
ZDT6	1.74793

IGD at D=30

Problem	MOGA
ZDT1	0.05305
ZDT2	0.09509
ZDT3	0.0448
ZDT4	12.6983
ZDT6	0.28662

4. Extended Comparison

GD at D=10

Problem	MOGA	MOGA Bonus	Crowding	Crowding + epsilon	Crowding + HV
ZDT1	0.22163	4.080e-04	7.407e-05	1.144e-04	5.601e-05
ZDT2	0.36434	3.989e-04	3.055e-05	4.958e-05	3.032e-05
ZDT3	0.19688	3.777e-04	5.480e-05	1.115e-04	5.435e-05
ZDT4	26.8802	0.06089	0.0568	0.07836	0.05578
ZDT6	1.56325	0.00377	2.708e-05	5.498e-05	2.709e-05

IGD at D=10

Problem	MOGA	MOGA Bonus	Crowding	Crowding + epsilon	Crowding + HV
ZDT1	0.01539	6.923e-05	2.403e-05	5.424e-05	2.031e-05
ZDT2	0.03393	7.841e-05	2.491e-05	5.878e-05	2.755e-05
ZDT3	0.01357	0.0017	1.377e-04	1.612e-04	1.377e-04
ZDT4	1.68026	0.07781	0.0725	0.07256	0.0725
ZDT6	0.20628	0.00158	1.538e-04	1.684e-04	1.538e-04

GD at D=30

Problem	MOGA	MOGA Bonus	Crowding	Crowding + epsilon	Crowding + HV
ZDT1	0.33038	6.610e-04	1.272e-04	1.798e-04	1.272e-04
ZDT2	0.7451	2.054e-04	1.525e-04	1.856e-04	1.525e-04
ZDT3	0.31074	3.957e-04	1.785e-04	2.033e-04	1.785e-04
ZDT4	104.99	1.31694	1.9144	1.96139	1.9144
ZDT6	1.74793	0.04988	0.04794	0.05079	0.04794

IGD at D=30

Problem	MOGA	MOGA Bonus	Crowding	Crowding + epsilon	Crowding + HV
ZDT1	0.05305	2.117e-04	1.273e-04	1.404e-04	1.273e-04
ZDT2	0.09509	2.283e-04	1.537e-04	1.617e-04	1.537e-04
ZDT3	0.0448	5.425e-04	2.526e-04	2.610e-04	2.526e-04
ZDT4	12.6983	0.98195	1.16444	1.16444	1.16444
ZDT6	0.28662	0.02195	0.02205	0.02205	0.02205

5. Visual Analysis

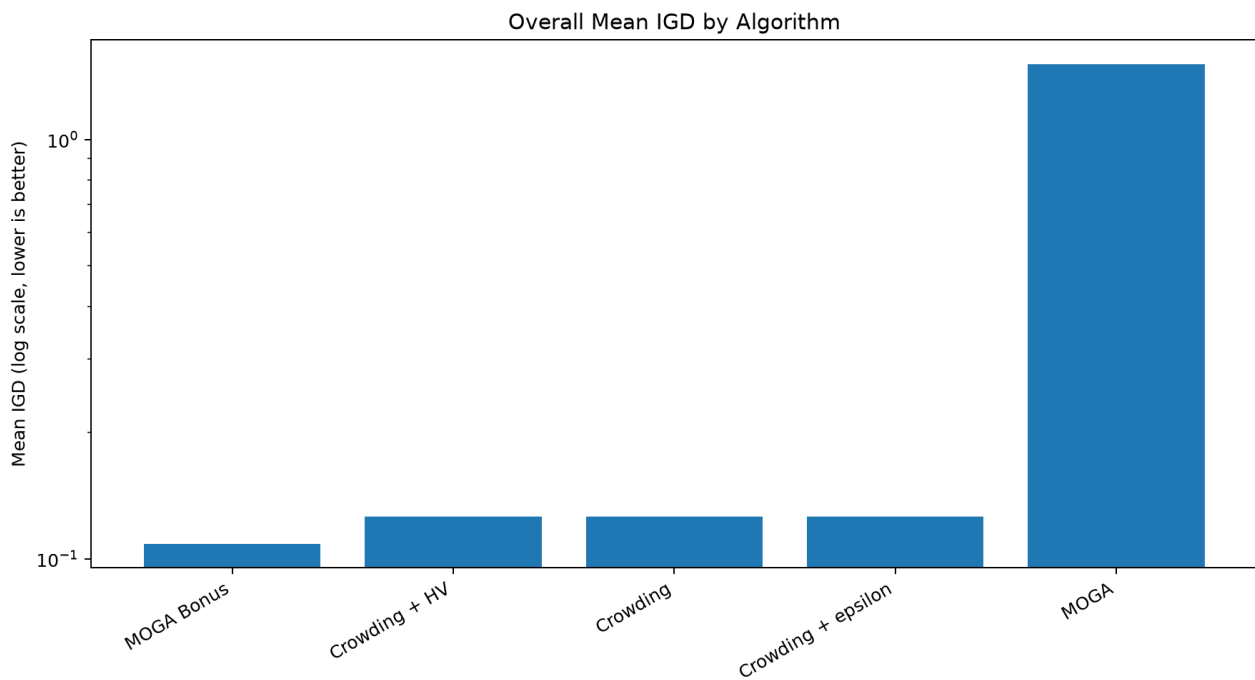


Figure 1. Overall mean IGD by algorithm. Lower values indicate better average Pareto-front coverage.

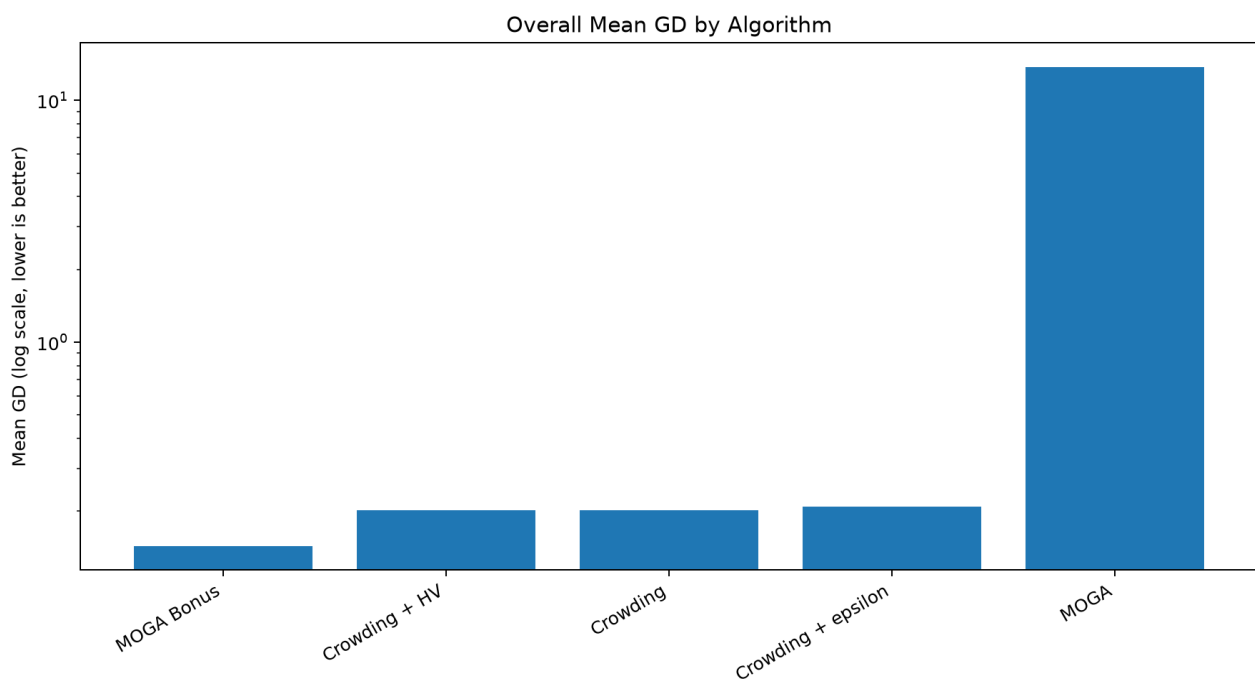


Figure 2. Overall mean GD by algorithm. Lower values indicate closer convergence to the reference front.

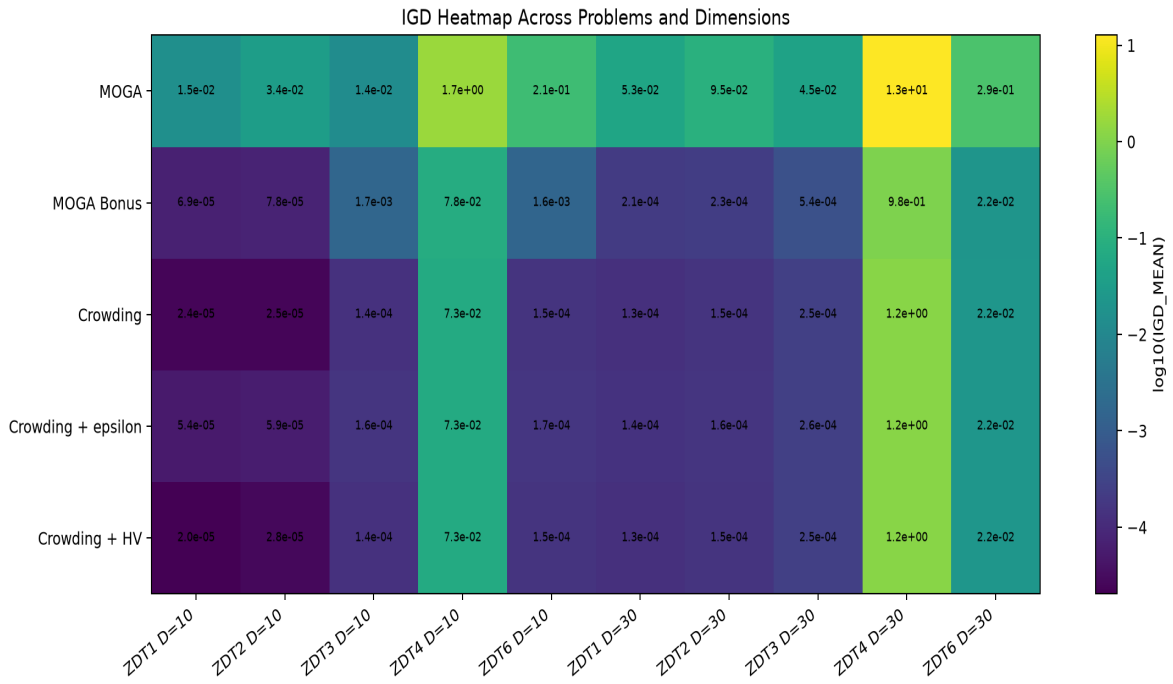


Figure 3. IGD heatmap across all problems and dimensions.

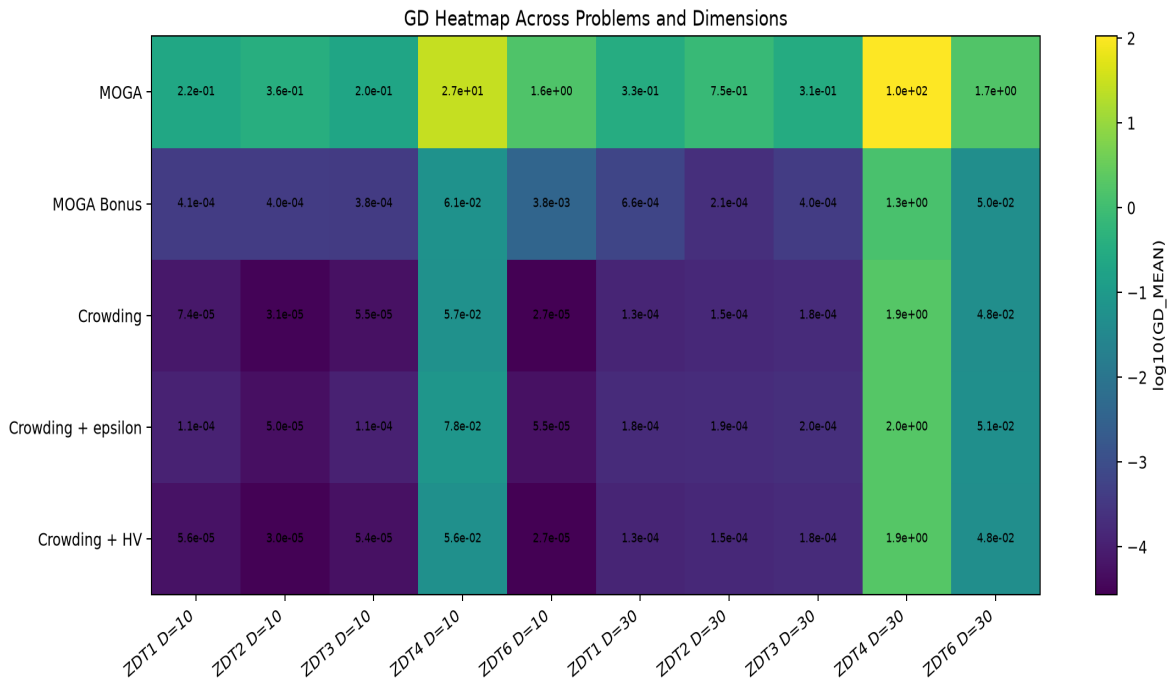


Figure 4. GD heatmap across all problems and dimensions. ZDT4 dominates the difficulty pattern.

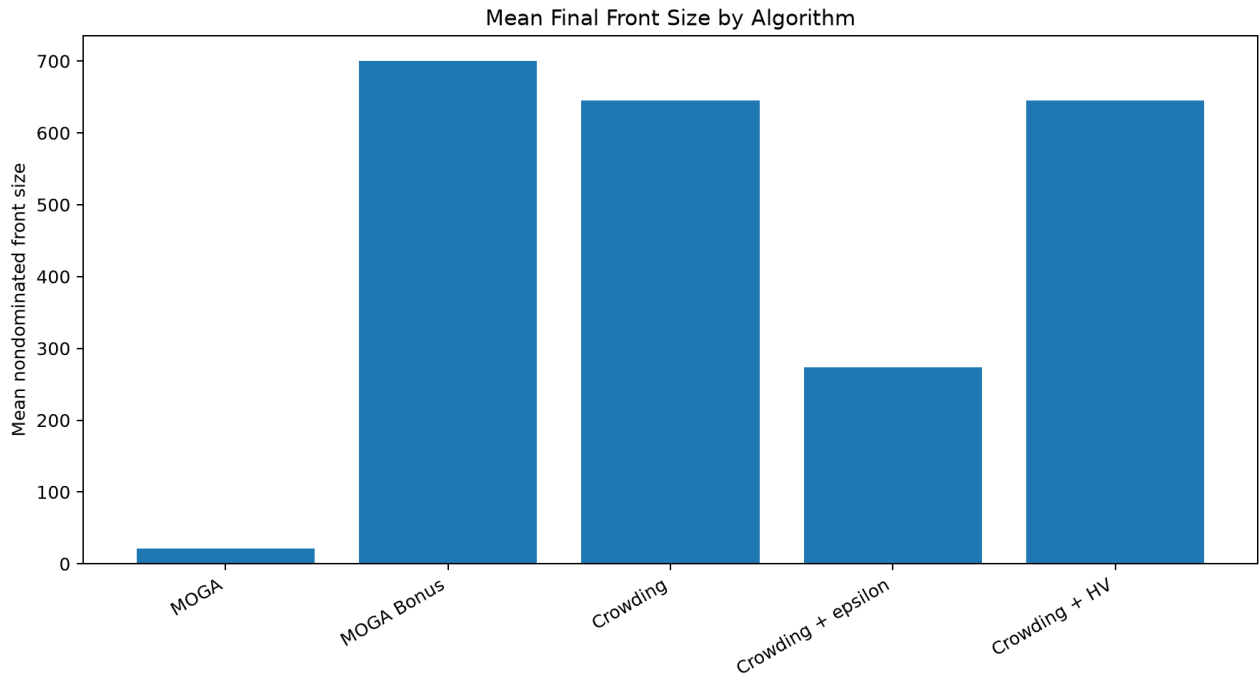


Figure 5. Mean final nondominated front size. Archive-based methods retain substantially more solutions than the baseline.

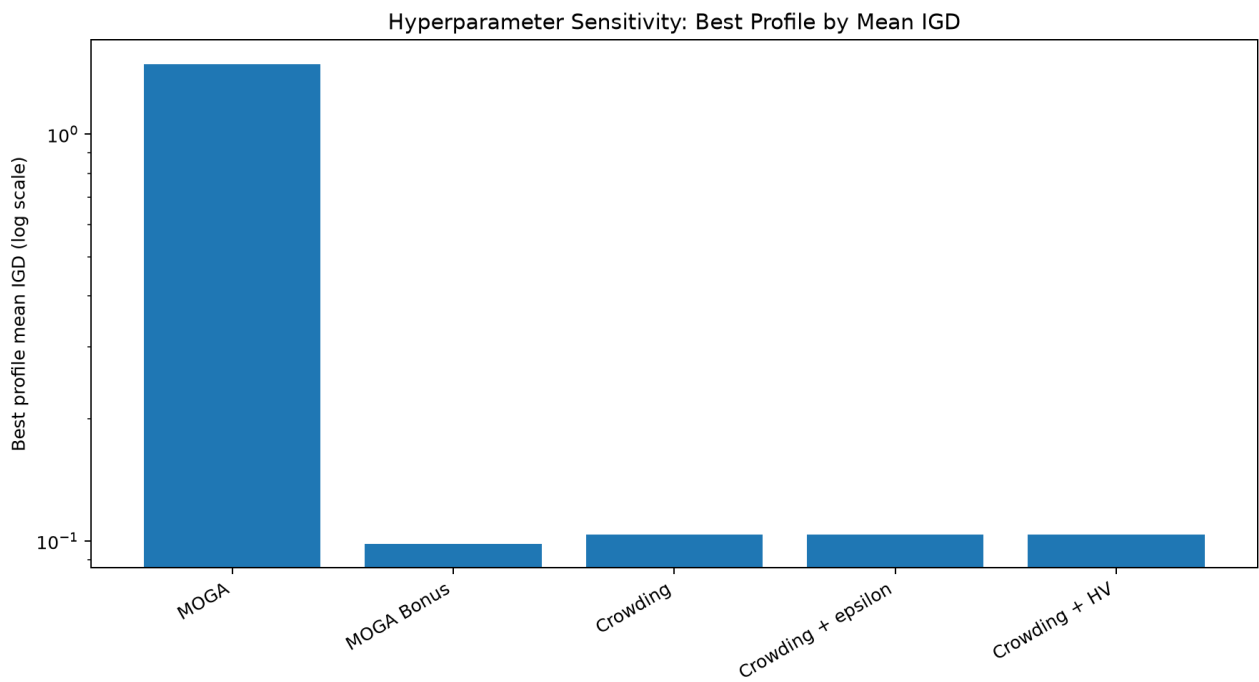


Figure 6. Hyperparameter sensitivity summary using the best tested profile by mean IGD for each algorithm.

6. Overall Ranking and Sensitivity

Rank	Algorithm	Mean IGD	Median IGD	Mean GD	Mean front size
1	MOGA Bonus	0.10861	0.00106	0.14339	699.84
2	Crowding + HV	0.12599	1.537e-04	0.20187	645.05
3	Crowding	0.12599	1.537e-04	0.20198	645.06
4	Crowding + epsilon	0.12601	1.650e-04	0.20914	274.07
5	MOGA	1.51273	0.07407	13.735	20.85

Algorithm	Best profile	Mean IGD	Mean GD
MOGA	mutation_low	1.48704	13.8074
MOGA Bonus	mutation_low	0.09849	0.13504
Crowding	mutation_low	0.10392	0.17692
Crowding + epsilon	mutation_low	0.10394	0.18436
Crowding + HV	mutation_low	0.10392	0.17684

7. Discussion

The baseline MOGA completes the benchmark protocol but produces much weaker GD and IGD than the archive-based variants. Its mean final front contains only about 20.85 nondominated solutions, compared with hundreds for the archive-based methods, which helps explain the poorer IGD and incomplete front coverage.

MOGA Bonus has the best overall mean IGD. Tournament selection provides more consistent selection pressure than roulette-style selection; elitism protects strong solutions; adaptive mutation balances exploration with refinement; and the external archive prevents nondominated solutions from disappearing between generations.

Crowding and Crowding + HV have especially strong typical performance. Crowding distance is a defensible diversity mechanism because it improves distribution without changing the optimization objective itself. Epsilon filtering reduces near-duplicates but can remove nearby points that still benefit IGD. Hypervolume-contribution pruning is more indicator-driven and can preserve high-value points, although its behavior depends on the reference point.

ZDT4 remains the dominant failure case. Its multimodal structure creates local optima away from the true Pareto front, so it exposes the limits of all tested variants more clearly than the easier ZDT cases.

8. Conclusion

Across the tested ZDT suite, the strongest overall mean IGD is achieved by MOGA Bonus, while crowding-based variants achieve very strong median IGD and diversity. Relative to the baseline aggregate mean IGD of 1.512730, MOGA Bonus reaches 0.108612, an improvement of approximately 92.8%. The results support archive and diversity mechanisms as substantial improvements, but they do not justify a universal claim that one variant dominates every benchmark.

Reproducibility

The repository contains the implementation, benchmark runners, tuning utilities, archived result tables, visualization generator, and this report. The figures and PDF are regenerated from the committed numeric results by GitHub Actions so the public artifact remains internally reproducible.